Reader: If the numerical function is a mapping of one numerical set onto another numerical set, then the operator can be considered as a mapping of a set of numerical function onto another set of functions, and the functional as a mapping of a set of functions onto a numerical set.

Author: You are quite right.

Reader: I have noticed that you persistently use the term "numerical function" (and I follow suit), but usually one simply says "function". Just how necessary is the word "numerical"?

Author: You have touched upon a very important aspect. The point is that in modern mathematics the concept of a function is substantially broader than the concept of a numerical function. As a matter of fact, the concept of a function includes, as particular cases, a numerical function as well as an operator and a functional, because the essence in all the three is a mapping of one set onto another independently of the nature of the sets. You have noticed that both operators and functionals are mappings of certain sets onto certain sets. In a particular case of mapping of a numerical set onto a numerical set we come to a numerical function. In a more general case, however, sets to be mapped can be arbitrary. Consider a few examples.

Example 1: Let D be a set of working days in an academic year, and E a set of students in a class. Using these sets, we can define a function realizing a schedule for the students on duty in the classroom. In compiling the schedule, each element of D (every working day in the year) is placed in one-to-one correspondence with a certain element of E (a certain student). This function is a mapping of the set of working days onto the set of students. We may add that the domain of the function consists of the working days and the range is defined by the set of the students.

Reader: It sounds a bit strange. Moreover, these sets have finite numbers of elements.

Author: This last feature is not principal.